

De Gouden Tarwe SQL Database Challenge

(Day 5 – Part 1)

MySQL

Problem Statement

De Gouden Tarwe (Dutch for The Golden Wheat) is a premium bakery chain based in the Netherlands. Established in 2018, the company has grown rapidly and now operates more than 30 branches across 20 major Dutch cities including Amsterdam, Rotterdam, Utrecht, The Hague, Eindhoven, and Groningen.

The bakery sells authentic Dutch baked goods from everyday bread like tijgerbrood and volkorenbrood, to traditional pastries like stroopwafels, appeltaart, bossche bollen, and seasonal specialties like oliebollen (only sold in December) and pepernoten (only sold during the Sinterklaas season in November).

The company employs over 450 people across all branches: bakers, assistant bakers, sales staff, branch managers, delivery drivers, and support staff. It serves tens of thousands of customers, many of whom participate in a loyalty program with tiered benefits (Bronze, Silver, Gold, Platinum).

The Current Problem

De Gouden Tarwe currently stores all its operational data in flat CSV files. As the business has grown, this approach has become unmanageable:

1. Data redundancy: The same product name, price, and description are repeated in every transaction record.
2. No formal relationships: There is no way to enforce that a branch referenced in an order actually exists.
3. No data integrity: A typo in a city name creates a new city that does not actually exist.
4. Impossible to answer business questions: Simple queries like What is our best-selling product in Rotterdam? require manually cross-referencing multiple files.
5. Performance problems: With millions of rows, opening and searching CSV files is painfully slow.

Progress – Day 5

On Day 05, i will answer this question using SQL queries:

1. What were top 10 best-selling products per branch last month?

To answer this question, we need orders, order_items, and products table. Orders for identifying orders with 'Vooltoid' status, order_items for identifying the product ordered, and products for identifying the product's name.

```
WITH product_sales AS (  
  SELECT  
    o.branch_id,  
    oi.product_id,  
    p.product_name,  
    SUM(oi.quantity) AS total_sold  
  FROM orders o  
  LEFT JOIN order_items oi  
    ON o.order_id = oi.order_id
```

```

LEFT JOIN products p
  ON oi.product_id = p.product_id
WHERE o.order_datetime >= '2023-12-01'
      AND o.order_datetime < '2024-01-01'
      AND o.order_status = 'Voltooid'
GROUP BY
  o.branch_id,
  oi.product_id,
  p.product_name)

SELECT
  branch_id,
  product_id,
  product_name,
  total_sold
FROM (
  SELECT *,
         ROW_NUMBER() OVER (
           PARTITION BY branch_id
           ORDER BY total_sold DESC
         ) AS ranking
    FROM product_sales
  ) ranked
WHERE ranking <= 10
ORDER BY branch_id, ranking;

```

And the results are:

- 1) Top 10 best-selling product from De Gouden Tarwe Amsterdam 1 are Volkoren Tarwebrood with 1055 units sold, Mergpijpje with 1007 units sold, Roombroodje with 1005 units sold, Gevulde Koek with 1004 units sold, Waldkorn Brood with 1001 units sold, Afbakbroodje and Jan Hagel each sold 1000, Moorkop with 999 units sold, Bruine Bol with 998 units sold, and Appleflap with 997 units sold.
- 2) Top 10 best-selling product from De Gouden Tarwe Amsterdam 2 are Roggebrood with 1150 units sold, Olijvenbrood with 1146 units sold, Petit Four Assortiment with 1134 units sold, Appeltart (punt) with 1108 units sold, Roomboterstaaf with 1106 units sold, Kretenbrood with 1105 units sold, Croissant Naturel with 1098 units sold, Afbakbroodje with 1096 units sold, Bruin Casino Brood with 1094 units sold, and Frikandelbroodje with 1093 units sold.
- 3) Top 10 best-selling product from De Gouden Tarwe Amsterdam 3 are Gevule Koek with 368 units sold, Cappucino with 345 units sold, Olijvenbrood with 344 units sold, Eierkoek with 342 units sold, Meerganen Brood with 340 units sold, Tompouce with 338 units sold, Mergpijpje with 335 units sold, Verse Jus d'Orange with 334 units sold, Waldkorn Brood with 332 units sold, and Bruine Bol with 325 units sold.
- 4) Top 10 best-selling product from De Gouden Tarwe Rotterdam 1 are Kaiserbroodje with 1202 units sold, Koffie with 1140 units sold, Roomboterstaaf with 1135 units sold, Latte Macchiato with 1128 units sold, Thee with 1124 units sold, Brabants Suikerbrood with 1123 units sold, Limburgse Vlaai Rijst with 1114 units sold, Kletskop with 1108 units sold, Appeltaart (heel) with 1086 units sold, and Frikandelbroodje with 1080 units sold
- 5) Top 10 best-selling product from De Gouden Tarwe Rotterdam 2 are Fris (cola/fanta/sinas) with 1124 units sold, Appeltaart (punt) with 1112 units sold, Oliebol Naturel with 1098 units sold, Moorkop with 1091 units sold, Limburgse

Vlaai Kers with 1079 units sold, Rozijnenbrood with 1069 units sold, Appeltaart (heel) with 1068 units sold, Pain au Chocolat with 1064 units sold, Pistolet with 1058 units sold, and Volkoren Tarwebrood with 1056 units sold.

- 6) Top 10 best-selling product from De Gouden Tarwe Rotterdam 3 are Kaiserbroodje with 716 units sold, Limburgse Vlaai Abrikoos with 703 units sold, Appeltaart (punt) with 688 units sold, Meergranen Brood with 683 units sold, Appelflap with 664 units sold, Chocolademelk with 663 units sold, Croissant Chocolate with 654 units sold, Appeltaart (heel) with 651 units sold, Water (Spa Blauw) with 650 units sold, and Speltbrood with 648 units sold.
- 7) Top 10 best-selling product from De Gouden Tarwe Den Haag 1 are Bonbons Doos (16 stuks) with 1124 units sold, Tompouce with 1103 units sold, Ciabatta with 1099 units sold, Haagse Kakker with 1098 units sold, Oliebol Naturel with 1074 units sold, Stroopwafel (vers) with 1073 units sold, Espresso with 1071 units sold, Gevulde Koek with 1061 units sold, Amandelstaaf with 1053 units sold, and Olijvenbrood with 1052 units sold.
- 8) Top 10 best-selling product from De Gouden Tarwe Den Haag 2 are Verse Muntthee with 739 units sold, Moorkop with 691 units sold, Speltbrood with 686 units sold, Waldkorn Brood with 680 units sold, Slagroomtaart with 679 units sold, Verse Jus d'Orange with 676 units sold, Stroopwafel (vers) with 668 units sold, Roomboterstaaf with 668 units sold, Volkoren Tarwebrood with 637 units sold, and Limburgse Vlaai Kers with 636 units sold.
- 9) Top 10 best-selling product from De Gouden Tarwe Den Haag 3 Puddingbroodje with 1169 units sold, Oliebol Rozijnen with 1157 units sold, Boterkoek with 1115 units sold, Rozijnenbol met Spijs with 1108 units sold, Saucijzenbroodje with 1105 units sold, Fris (cola/fanta/sinas) with 1097 units sold, Bruine Bol with 1095 units sold, Roggebrood with 1092 units sold, Brabants Suikerbrood with 1090 units sold, and Latte Macchiato with 1089 units sold.
- 10) Top 10 best-selling product from De Gouden Tarwe Utrecht 1 are Moorkop with 1099 units sold, Pistolet with 1096 units sold, Eierkoek with 1086 units sold, Cappuccino with 1076 units sold, Appeltaart (punt) with 1074 units sold, Kletskop with 1069 units sold, Oliebol Rozijnen with 1065 units sold, Haagse Kakker with 1058 units sold, Croissant Naturel with 1050 units sold, and Gevulde Koek with 1050 units sold.
- 11) Top 10 best-selling product from De Gouden Tarwe Utrecht 2 are Limburgse Vlaai Kers with 1225 units sold, Chocolademelk with 1158 units sold, Rozijnenbrood with 1135 units sold, Zuurdesembrood with 1129 units sold, Brabants Suikerbrood with 1120 units sold, Slagroomtaart with 1107 units sold, Olijvenbrood with 1103 units sold, Roggebrood with 1089 units sold, Oliebol Naturel with 1078 units sold, and Speltbrood with 1075 units sold.
- 12) Top 10 best-selling product from De Gouden Tarwe Utrecht 3 are Fries Suikerbrood with 382 units sold, Cappuccino with 354 units sold, Croissant Naturel with 343 units sold, Oliebol Naturel with 337 units sold, Verse Muntthee with 336 units sold, Appelflap with 334 units sold, Croissant Chocolate with 333 units sold, Stroopwafel (vers) with 333 units sold, Amandelstaaf with 332 units sold, and Limburgse Vlaai Rijst with 330 units sold.

- 13) Top 10 best-selling product from De Gouden Tarwe Eindhoven 1 are Zuurdesembrood with 1134 units sold, Banketstaaf with 1124 units sold, Arnhemse Meisjes with 1122 units sold, Bossche Bol with 1115 units sold, Haagse Kakker with 1105 units sold, Verse Muntthee with 1104 units sold, Bonbons Doos (9 stuks) with 1092 units sold, Wit Puntje with 1088 units sold, Jan Hagel with 1083 units sold, and Bruin Casino Brood with 1080 units sold.
- 14) Top 10 best-selling product from De Gouden Tarwe Eindhoven 2 are Appeltaart (punt) with 366 units sold, Espresso with 356 units sold, Speltbrood with 351 units sold, Kaiserbroodje with 349 units sold, Waldkorn Brood with 333 units sold, Bruine Bol with 333 units sold, Kletskop with 333 units sold, Bonbons Doos (16 stuks) with 324 units sold, Ciabatta with 315 units sold, and Water (Spa Blauw) with 314 units sold.
- 15) Top 10 best-selling product from De Gouden Tarwe Eindhoven 3 are Appeltaart (heel) with 362 units sold, Pain au Chocolat with 354 units sold, Gevulde Koek with 349 units sold, Boterkoek with 347 units sold, Kerststol with 345 units sold, Cappuccino with 337 units sold, Oliebol Naturel with 334 units sold, Gevulde Speculaas with 333 units sold, Croissant Naturel with 327 units sold, and Moorkop with 327 units sold.
- 16) Top 10 best-selling product from De Gouden Tarwe Groningen Top 10 best-selling product from De Gouden Tarwe 3 are Croissant Naturel with 1150 units sold, Amandelstaaf with 1140 units sold, Arnhemse Meisjes with 1131 units sold, Verse Muntthee with 1121 units sold, Limburgse Vlaai Abrikoos with 1114 units sold, Bossche Bol with 1111 units sold, Fries Suikerbrood with 1104 units sold, Pain au Chocolat with 1102 units sold, Fris (cola/fanta/sinas) with 1100 units sold, and Bruin Casino Brood with 1097 units sold.
- 17) Top 10 best-selling product from De Gouden Tarwe Tilburg 1 are Bonbons Doos (9 stuks) with 1136 units sold, Maisbroodje with 1116 units sold, Appeltaart (heel) with 1111 units sold, Puddingbroodje with 1091 units sold, Cappuccino with 1085 units sold, Oliebol Naturel with 1082 units sold, Pain au Chocolat with 1072 units sold, Koffie with 1063 units sold, Rozijnenbol met Spijs with 1059 units sold, and Volkoren Tarwebrood with 1056 units sold.
- 18) Top 10 best-selling product from De Gouden Tarwe Tilburg 2 are Croissant Amandel with 1174 units sold, Thee with 1161 units sold, Croissant Naturel with 1114 units sold, Meergranen Brood with 1089 units sold, Bonbons Doos (16 stuks) with 1089 units sold, Jan Hagel with 1086 units sold, Brabants Suikerbrood with 1081 units sold, Kaiserbroodje with 1078 units sold, Limburgse Vlaai Abrikoos with 1077 units sold, and Boterkoek with 1076 units sold.
- 19) Top 10 best-selling product from De Gouden Tarwe Almere 1 are Oliebol Rozijnen with 1079 units sold, Meergranen Brood with 1063 units sold, Croissant Amandel with 1045 units sold, Pistolet with 1037 units sold, Waldkorn Brood with 1036 units sold, Volkoren Tarwebrood with 1030 units sold, Worstenbroodje with 1028 units sold, Croissant Naturel with 1027 units sold, Appeltaart (punt) with 1026 units sold, and Kletskop with 1021 units sold.
- 20) Top 10 best-selling product from De Gouden Tarwe Almere 2 are Appeltaart (heel) with 394 units sold, Pain au Chocolat with 370 units sold, Moorkop with 358 units sold, Amandelstaaf with 358 units sold, Brabants Suikerbrood with 354 units sold,

Frikandelbroodje with 353 units sold, Maisbroodje with 352 units sold, Limburgse Vlaai Abrikoos with 351 units sold, Haagse Kakker with 346 units sold, and Bruine Bol with 340 units sold.

- 21) Top 10 best-selling product from De Gouden Tarwe Breda are Roombroodje with 391 units sold, Croissant Amandel with 363 units sold, Volkoren Tarwebrood with 358 units sold, Meergranen Brood with 357 units sold, Roggebrood with 357 units sold, Frikandelbroodje with 357 units sold, Pain au Chocolat with 353 units sold, Tompouce with 347 units sold, Cappuccino with 345 units sold, and Limburgse Vlaai Rijst with 344 units sold.
- 22) Top 10 best-selling product from De Gouden Tarwe Nijmegen are Olijvenbrood with 404 units sold, Verse Muntthee with 379 units sold, Chocolademelk with 377 units sold, Pompoenpitten Brood with 370 units sold, Puddingbroodje with 356 units sold, Fries Suikerbrood with 350 units sold, Bruine Bol with 342 units sold, Saucijzenbroodje with 341 units sold, Fris (cola/fanta/sinas) with 340 units sold, and Limburgse Vlaai Abrikoos with 339 units sold.
- 23) Top 10 best-selling product from De Gouden Tarwe Haarlem are Fris (cola/fanta/sinas) with 682 units sold, Rozijnenbrood with 676 units sold, Croissant Amandel with 675 units sold, Wit Puntje with 674 units sold, Gevulde Speculaas with 672 units sold, Speltbrood with 669 units sold, Wit Tijgerbrood with 658 units sold, Boterkoek with 657 units sold, Stroopwafel (vers) with 653 units sold, and Volkoren Tarwebrood with 651 units sold.
- 24) Top 10 best-selling product from De Gouden Tarwe Arnhem 1 are Pistolet with 378 units sold, Boterkoek with 348 units sold, Bruine Bol with 345 units sold, Oliebol Naturel with 333 units sold, Worstebroodje with 329 units sold, Fris (cola/fanta/sinas) with 329 units sold, Fries Suikerbrood with 328 units sold, Croissant Naturel with 328 units sold, Meergranen Brood with 322 units sold, and Chocolademelk with 322 units sold.
- 25) Top 10 best-selling product from De Gouden Tarwe Arnhem 2 are Mergpijpje with 396 units sold, Boterkoek with 378 units sold, Koffiebroodje with 360 units sold, Fries Suikerbrood with 358 units sold, Verse Jus d'Orange with 348 units sold, Volkoren Tarwebrood with 344 units sold, Tompouce with 339 units sold, Rozijnenbrood with 336 units sold, Bonbons Doos (9 stuks) with 336 units sold, and Gevulde Speculaas with 334 units sold.
- 26) Top 10 best-selling product from De Gouden Tarwe Leiden 1 are Haagse Kakker with 373 units sold, Brabants Suikerbrood with 349 units sold, Moorkop with 339 units sold, Fris (cola/fanta/sinas) with 339 units sold, Wit Tijgerbrood with 332 units sold, Koffiebroodje with 331 units sold, Wit Puntje with 322 units sold, Roombroodje with 317 units sold, Bonbons Doos (16 stuks) with 317 units sold, and Bonbons Doos (9 stuks) with 311 units sold.
- 27) Top 10 best-selling product from De Gouden Tarwe Leiden 2 are Wit Casino Brood with 416 units sold, Puddingbroodje with 406 units sold, Frikandelbroodje with 389 units sold, Saucijzenbroodje with 381 units sold, Appelflap with 375 units sold, Chocolademelk with 375 units sold, Tompouce with 370 units sold, Petit Four Assortiment with 365 units sold, Verse Muntthee with 363 units sold, and Ciabatta with 359 units sold.

- 28) Top 10 best-selling product from De Gouden Tarwe Maastricht are Olijvenbrood with 734 units sold, Haagse Kakker with 712 units sold, Fries Suikerbrood with 704 units sold, Maisbroodje with 696 units sold, Volkoren Tarwebrood with 688 units sold, Gevulde Speculaas with 687 units sold, Kerststol with 678 units sold, Krentenbrood with 671 units sold, Brabants Suikerbrood with 663 units sold, and Volkoren Bol with 662 units sold.
- 29) Top 10 best-selling product from De Gouden Tarwe Delft are Speltbrood with 378 units sold, Croissant Naturel with 369 units sold, Thee with 369 units sold, Fries Suikerbrood with 362 units sold, Bruine Bol with 355 units sold, Volkoren Bol with 355 units sold, Boterkoek with 349 units sold, Waldkorn Brood with 345 units sold, Croissant Chocolate with 344 units sold, and Worstenbroodje with 343 units sold.
- 30) Top 10 best-selling product from De Gouden Tarwe Amersfoort 1 are Moorkop with 661 units sold, Latte Macchiato with 659 units sold, Speltbrood with 658 units sold, Croissant Naturel with 652 units sold, Boterkoek with 643 units sold, Bruin Casino Brood with 632 units sold, Appelflap with 632 units sold, Volkoren Tarwebrood with 627 units sold, Brabants Suikerbrood with 627 units sold, and Limburgse Vlaai Kers with 624 units sold.
- 31) Top 10 best-selling product from De Gouden Tarwe Amersfoort 2 are Rozijnenbol met Spijs with 357 units sold, Oliebol Naturel with 356 units sold, Pistolet with 355 units sold, Fries Suikerbrood with 349 units sold, Speculaas with 337 units sold, Petit Four Assortiment with 335 units sold, Puddingbroodje with 334 units sold, Bossche Bol with 330 units sold, Saucijzenbroodje with 327 units sold, and Krentenbrood with 324 units sold.
- 32) Top 10 best-selling product from De Gouden Tarwe Zwolle are Bonbons Doos (9 stuks) with 1139 units sold, Wit Casino Brood with 1106 units sold, Petit Four Assortiment with 1086 units sold, Jan Hagel with 1077 units sold, Water (Spa Blauw) with 1065 units sold, Bruin Casino Brood with 1051 units sold, Krentenbrood with 1046 units sold, Appelflap with 1041 units sold, Verse Jus d'Orange with 1035 units sold, and Latte Macchiato with 1028 units sold.
- 33) Top 10 best-selling product from De Gouden Tarwe Leeuwarden are Puddingbroodje with 1065 units sold, Petit Four Assortiment with 1044 units sold, Moorkop with 1037 units sold, Bonbons Doos (16 stuks) with 1036 units sold, Maisbroodje with 1022 units sold, Thee with 1021 units sold, Kaiserbroodje with 1020 units sold, Roggebrood with 1016 units sold, Cappuccino with 1009 units sold, and Limburgse Vlaai Abrikoos with 1001 units sold.
- 34) Top 10 best-selling product from De Gouden Tarwe Bosch are Oliebol Rozijnen with 748 units sold, Saucijzenbroodje with 691 units sold, Moorkop with 689 units sold, Fris (cola/fanta/sinas) with 674 units sold, Eierkoek with 673 units sold, Bonbons Doos (16 stuks) with 672 units sold, Roggebrood with 670 units sold, Tompouce with 669 units sold, Croissant Naturel with 667 units sold, and Afbakbroodje with 667 units sold.
- 35) Top 10 best-selling product from De Gouden Tarwe Dordrecht are Waldkorn Brood with 365 units sold, Eierkoek with 358 units sold, Amandelstaaf with 354 units sold, Kletskop with 348 units sold, Pompoenpitten Brood with 343 units sold, Speculaas with 342 units sold, Oliebol Rozijnen with 330 units sold, Verse

Muntthee with 321 units sold, Boterkoek with 320 units sold, and Meergranen Brood with 318 units sold.

From 35 branch, the most best-selling product is Croissant Naturel with 8125 units sold.

2. What is the revenue per city per quarter, with year-over-year growth?

To answer this question, we need the orders, branches, and cities tables. The orders table is used to filter transactions with 'Voltooid' status, the branches table is used to map each order to its corresponding branch, and the cities table is used to identify the city name for each branch.

```
USE de_gouden_tarwe;
WITH quarterly_revenue AS (
    SELECT
        c.city_name,
        YEAR(o.order_datetime) AS year,
        QUARTER(o.order_datetime) AS quarter,
        SUM(o.total_amount) AS revenue
    FROM orders o
    LEFT JOIN branches b
        ON o.branch_id = b.branch_id
    LEFT JOIN cities c
        ON b.city_id = c.city_id
    WHERE o.order_status = 'Voltooid'
    GROUP BY
        c.city_name,
        YEAR(o.order_datetime),
        QUARTER(o.order_datetime)
)

SELECT
    city_name,
    year,
    quarter,
    revenue,
    LAG(revenue) OVER (
        PARTITION BY city_name, quarter
        ORDER BY year
    ) AS last_year_revenue,

    ROUND(
        (revenue - LAG(revenue) OVER (
            PARTITION BY city_name, quarter
            ORDER BY year
        )) /
        LAG(revenue) OVER (
            PARTITION BY city_name, quarter
            ORDER BY year
        ) * 100, 2
    ) AS yoy_growth_percent

FROM quarterly_revenue
ORDER BY city_name, year, quarter;
```

The query result consists of 400 rows of data across six columns: city_name, year, quarter, revenue, last_year_revenue, and yoy_growth_percent. Here is the summary table showing the average YoY growth and its variability for each city. We'll use this to draw insights in the next part.

City name	Average YoY growth	YoY standard deviation
Utrecht	0.005000	1.470333
Rotterdam	-0.211250	1.616914
Amersfoort	-0.186250	1.721181
Den Haag	-0.080625	1.864258
Eindhoven	-0.158750	1.918996
Leiden	-0.191875	1.956239
Almere	0.450000	2.060984
Tilburg	-0.239375	2.063839
Amsterdam	0.013750	2.204050
Maastricht	0.080000	2.287109
Zwolle	-0.386250	2.378854
Den Bosch	0.140000	2.482821
Haarlem	0.071875	2.509399
Arnhem	-0.297500	2.759400
Leeuwarden	-0.732500	2.808246
Nijmegen	-0.356875	2.972417
Delft	0.728750	2.975341
Groningen	0.502500	3.092256
Breda	0.148750	3.715858
Dordrecht	0.060000	3.930946

Based on this table, key insights can be identified, including high-growth cities, declining cities, relatively stable cities, and highly volatile cities. These classifications are derived from the mean (average) and standard deviation of the data. The detailed results are as follows:

1) High-growth cities (strong positive growth)

Delft, Groningen, and Almere are classified as high-growth cities due to their positive average YoY growth, indicating strong revenue performance compared to the previous year.

2) Declining cities (negative growth)

Leeuwarden, Zwolle, Nijmegen are classified as declining cities due to their negative average YoY growth, indicating a downward trend in revenue compared to the previous year.

3) Relatively stable cities

Utrecht and Rotterdam are considered relatively stable cities due to their low standard deviation, indicating that their revenue performance is more consistent over time despite modest or negative growth.

4) Highly fluctuating cities

Dordrecht, Breda, and Groningen are classified as highly fluctuating cities due to their high standard deviation, indicating significant variations in revenue growth across quarters and less consistent performance.

3. Which ingredients need to be reordered (current stock below minimum)?

To answer this question, we need the orders, branches, and cities tables. The orders table is used to filter transactions with 'Voltooid' status, the branches table is used to map each order to its corresponding branch, and the cities table is used to identify the city name for each branch.

```
WITH ingredients_to_reorder AS (
  SELECT
```

```

        i.ingredient_id,
        i.ingredient_name,
        i_s.branch_id,
        b.branch_name,
        i_s.current_quantity,
        i_s.minimum_quantity,
        i_s.unit_of_measure,
        i.reorder_quantity,
        (i_s.minimum_quantity - i_s.current_quantity) AS stock_gap
FROM inventory_stock i_s
LEFT JOIN ingredients i
    ON i_s.ingredient_id = i.ingredient_id
LEFT JOIN branches b
    ON i_s.branch_id = b.branch_id
WHERE i_s.current_quantity < i_s.minimum_quantity
    AND i.is_active = TRUE
)

SELECT
    ingredient_id,
    ingredient_name,
    branch_id,
    branch_name,
    current_quantity,
    minimum_quantity,
    stock_gap,
    reorder_quantity,
    unit_of_measure
FROM ingredients_to_reorder
ORDER BY branch_id, ingredient_name;

```

A total of 48 ingredients need to be reordered, including details of which branch requires each item and in what quantity.

- 1) Koffie bonen at the De Gouden Tarwe Amsterdam 1 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 5,56 kg, while the minimum required level is 5,64 kg, resulting in a shortage of 0,08 kg. Based on the data, the recommended reorder quantity is 242 kg.
- 2) Paneermeel at the De Gouden Tarwe Amsterdam 1 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 9,67 kg, while the minimum required level is 11,21 kg, resulting in a shortage of 1,54 kg. Based on the data, the recommended reorder quantity is 338 kg.
- 3) Eiren (doos 30) at the De Gouden Tarwe Amsterdam 2 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 12,16 doos, while the minimum required level is 19,77 doos, resulting in a shortage of 7,61 doos. Based on the data, the recommended reorder quantity is 249 doos.
- 4) Pompoenpitten at the De Gouden Tarwe Amsterdam 3 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 9,96 kg, while the minimum required level is 10,49 kg, resulting in a shortage of 0,53 kg. Based on the data, the recommended reorder quantity is 448 kg.
- 5) Verse Munt at the De Gouden Tarwe Amsterdam 3 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current

stock level is 9,33 bos, while the minimum required level is 19,16 bos, resulting in a shortage of 1,9,83 bos. Based on the data, the recommended reorder quantity is 403 bos.

- 6) Basterdsuiker bruin at the De Gouden Tarwe Rotterdam 2 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 6,16 kg, while the minimum required level is 11,45 kg, resulting in a shortage of 5,29 kg. Based on the data, the recommended reorder quantity is 294 kg.
- 7) Poedersuiker at the De Gouden Tarwe Rotterdam 3 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 6,94 kg, while the minimum required level is 15,19 kg, resulting in a shortage of 8,25 kg. Based on the data, the recommended reorder quantity is 202 kg.
- 8) Abrikozen at the De Gouden Tarwe Den Haag 1 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 13,53 kg, while the minimum required level is 15,23 kg, resulting in a shortage of 1,70 kg. Based on the data, the recommended reorder quantity is 382 kg.
- 9) Nootmuskaat at the De Gouden Tarwe Den Haag 1 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 9,75 kg, while the minimum required level is 17,01 kg, resulting in a shortage of 7,26 kg. Based on the data, the recommended reorder quantity is 389 kg.
- 10) Thee (diverse) at the De Gouden Tarwe Den Haag 1 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 12,26 kg, while the minimum required level is 12,47 kg, resulting in a shortage of 0,21 kg. Based on the data, the recommended reorder quantity is 151 kg.
- 11) Amandelspijs at the De Gouden Tarwe Utrecht 1 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 5,38 kg, while the minimum required level is 8,69 kg, resulting in a shortage of 3,31 kg. Based on the data, the recommended reorder quantity is 397 kg.
- 12) Gist (droog) at the De Gouden Tarwe Utrecht 1 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 9,12 kg, while the minimum required level is 11,70 kg, resulting in a shortage of 2,58 kg. Based on the data, the recommended reorder quantity is 155 kg.
- 13) Koffie bonen at the De Gouden Tarwe Utrecht 2 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 9,31 kg, while the minimum required level is 13,17 kg, resulting in a shortage of 3,86 kg. Based on the data, the recommended reorder quantity is 242 kg.
- 14) Abrikozen at the De Gouden Tarwe Utrecht 3 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 5,79 kg, while the minimum required level is 5,89 kg, resulting in a

shortage of 0,10 kg. Based on the data, the recommended reorder quantity is 382 kg.

- 15) Appels (Elstar) at the De Gouden Tarwe Eindhoven 1 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 7,55 kg, while the minimum required level is 18,63 kg, resulting in a shortage of 11,08 kg. Based on the data, the recommended reorder quantity is 59 kg.
- 16) Appels (Elstar) at the De Gouden Tarwe Eindhoven 1 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 7,55 kg, while the minimum required level is 18,63 kg, resulting in a shortage of 11,08 kg. Based on the data, the recommended reorder quantity is 59 kg.
- 17) Volkorenmeel at the De Gouden Tarwe Eindhoven 1 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 10,64 kg, while the minimum required level is 17,96 kg, resulting in a shortage of 7,32 kg. Based on the data, the recommended reorder quantity is 405 kg.
- 18) Appels (Elstar) at the De Gouden Tarwe Eindhoven 2 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 13,39 kg, while the minimum required level is 17,89 kg, resulting in a shortage of 4,50 kg. Based on the data, the recommended reorder quantity is 59 kg.
- 19) Zonnebloempitten at the De Gouden Tarwe Eindhoven 2 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 14,51 kg, while the minimum required level is 17,31 kg, resulting in a shortage of 2,80 kg. Based on the data, the recommended reorder quantity is 105 kg.
- 20) Gehakt (saucijzenbroodjes) at the De Gouden Tarwe Groningen branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 10,48 kg, while the minimum required level is 17,33 kg, resulting in a shortage of 6,85 kg. Based on the data, the recommended reorder quantity is 408 kg.
- 21) Kristalsuiker at the De Gouden Tarwe Groningen branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 8,79 kg, while the minimum required level is 11,82 kg, resulting in a shortage of 3,03 kg. Based on the data, the recommended reorder quantity is 60 kg.
- 22) Kruidnagel (gemalen) at the De Gouden Tarwe Groningen branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 5,11 kg, while the minimum required level is 14,45 kg, resulting in a shortage of 9,34 kg. Based on the data, the recommended reorder quantity is 251 kg.
- 23) Bakpoeder at the De Gouden Tarwe Tilburg 1 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 5,31 kg, while the minimum required level is 9,43 kg, resulting in a shortage of 4,12 kg. Based on the data, the recommended reorder quantity is 379 kg.

- 24) Roomboter at the De Gouden Tarwe Tilburg 1 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 8,05 kg, while the minimum required level is 11,11 kg, resulting in a shortage of 3,06 kg. Based on the data, the recommended reorder quantity is 325 kg.
- 25) Zonnebloempitten at the De Gouden Tarwe Tilburg 2 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 6,48 kg, while the minimum required level is 11,78 kg, resulting in a shortage of 5,30 kg. Based on the data, the recommended reorder quantity is 105 kg.
- 26) Gist (droog) at the De Gouden Tarwe Almere 1 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 8,82 kg, while the minimum required level is 16,13 kg, resulting in a shortage of 7,31 kg. Based on the data, the recommended reorder quantity is 155 kg.
- 27) Thee (diverse) at the De Gouden Tarwe Breda branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 11,46 kg, while the minimum required level is 13,62 kg, resulting in a shortage of 2,16 kg. Based on the data, the recommended reorder quantity is 151 kg.
- 28) Thee (diverse) at the De Gouden Tarwe Nijmegen branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 17,27 kg, while the minimum required level is 18,45 kg, resulting in a shortage of 1,18 kg. Based on the data, the recommended reorder quantity is 151 kg.
- 29) Zonnebloempitten at the De Gouden Tarwe Nijmegen branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 7,10 kg, while the minimum required level is 12,90 kg, resulting in a shortage of 5,80 kg. Based on the data, the recommended reorder quantity is 105 kg.
- 30) Lijnzaad at the De Gouden Tarwe Haarlem branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 6,19 kg, while the minimum required level is 19,63 kg, resulting in a shortage of 13,44 kg. Based on the data, the recommended reorder quantity is 209 kg.
- 31) Margarine at the De Gouden Tarwe Haarlem branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 6,31 kg, while the minimum required level is 12,60 kg, resulting in a shortage of 6,29 kg. Based on the data, the recommended reorder quantity is 459 kg.
- 32) Pompoenpitten at the De Gouden Tarwe Haarlem branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 15,11 kg, while the minimum required level is 17,63 kg, resulting in a shortage of 2,52 kg. Based on the data, the recommended reorder quantity is 448 kg.
- 33) Nootmuskaat at the De Gouden Tarwe Arnhem 1 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current

stock level is 6,57 kg, while the minimum required level is 12,10 kg, resulting in a shortage of 5,53 kg. Based on the data, the recommended reorder quantity is 389 kg.

- 34) Paneermeel at the De Gouden Tarwe Arnhem 1 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 10,57 kg, while the minimum required level is 11,33 kg, resulting in a shortage of 0,76 kg. Based on the data, the recommended reorder quantity is 338 kg.
- 35) Koffie bonen at the De Gouden Tarwe Arnhem 2 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 13,97 kg, while the minimum required level is 19,63 kg, resulting in a shortage of 5,66 kg. Based on the data, the recommended reorder quantity is 242 kg.
- 36) Roggemeel at the De Gouden Tarwe Arnhem 2 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 8,35 kg, while the minimum required level is 11,48 kg, resulting in a shortage of 3,13 kg. Based on the data, the recommended reorder quantity is 159 kg.
- 37) Rozijnen at the De Gouden Tarwe Leiden 1 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 5,39 kg, while the minimum required level is 7,29 kg, resulting in a shortage of 1,90 kg. Based on the data, the recommended reorder quantity is 217 kg.
- 38) Zonnebloempitten at the De Gouden Tarwe Leiden 1 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 5,61 kg, while the minimum required level is 15,76 kg, resulting in a shortage of 10,15 kg. Based on the data, the recommended reorder quantity is 105 kg.
- 39) Appels (Elstar) at the De Gouden Tarwe Leiden 2 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 6,10 kg, while the minimum required level is 6,11 kg, resulting in a shortage of 0,01 kg. Based on the data, the recommended reorder quantity is 59 kg.
- 40) Margarine at the De Gouden Tarwe Maastricht branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 8,37 kg, while the minimum required level is 18,78 kg, resulting in a shortage of 10,41 kg. Based on the data, the recommended reorder quantity is 459 kg.
- 41) Amandelspijs at the De Gouden Tarwe Delft branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 13,90 kg, while the minimum required level is 19,30 kg, resulting in a shortage of 5,40 kg. Based on the data, the recommended reorder quantity is 397 kg.
- 42) Vanillesuiker at the De Gouden Tarwe Delft branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 5,34 kg, while the minimum required level is 7,62 kg, resulting in a

shortage of 2,28 kg. Based on the data, the recommended reorder quantity is 480 kg.

- 43) Zonnebloemolie at the De Gouden Tarwe Delft branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 8,82 liter, while the minimum required level is 16,96 liter, resulting in a shortage of 8,14 liter. Based on the data, the recommended reorder quantity is 128 liter.
- 44) Pompoenpitten at the De Gouden Tarwe Amersfoort 2 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 5,22 kg, while the minimum required level is 9,70 kg, resulting in a shortage of 4,48 kg. Based on the data, the recommended reorder quantity is 448 kg.
- 45) Stroop (donker) at the De Gouden Tarwe Amersfoort 2 branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 13,77 kg, while the minimum required level is 14,34 kg, resulting in a shortage of 0,57 kg. Based on the data, the recommended reorder quantity is 152 kg.
- 46) Abrikozen at the De Gouden Tarwe Zwolle branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 6,87 kg, while the minimum required level is 10,57 kg, resulting in a shortage of 3,70 kg. Based on the data, the recommended reorder quantity is 382 kg.
- 47) Sinaasappels (pers) at the De Gouden Tarwe Zwolle branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 17,67 kg, while the minimum required level is 18,94 kg, resulting in a shortage of 1,27 kg. Based on the data, the recommended reorder quantity is 102 kg.
- 48) Succade at the De Gouden Tarwe Zwolle branch requires attention for restocking, as the current stock is approaching the minimum threshold. The current stock level is 7,53 kg, while the minimum required level is 12,70 kg, resulting in a shortage of 5,17 kg. Based on the data, the recommended reorder quantity is 477 kg.

4. What is the average revenue per employee?

To answer this question, we use the employees table and the summary_employee_performance table. The employees table is used to retrieve employee details such as name and position, while the summary_employee_performance table is used to provide pre-aggregated performance metrics such as total revenue, total working hours, and number of sales days. This summary table is used instead of raw transactional data to improve query performance, since the aggregation process has already been done beforehand.

```
SELECT
    e.employee_id,
    CONCAT(e.first_name, ' ', e.last_name) AS employee_name,
    e.position_title,
    SUM(s.total_revenue) AS total_revenue_all_time,

    SUM(s.total_hours_worked) AS total_hours_worked,

    ROUND(SUM(s.total_revenue) / NULLIF(SUM(s.total_hours_worked),
0), 2) AS avg_revenue_per_hour,
```

```

ROUND(SUM(s.total_revenue) / NULLIF(SUM(s.days_with_sales), 0),
2) AS avg_revenue_per_sale_day
FROM employees e
JOIN summary_employee_performance s ON e.employee_id = s.employee_id
WHERE e.is_active = TRUE
GROUP BY e.employee_id, e.first_name, e.last_name, e.position_title
ORDER BY avg_revenue_per_hour DESC;

```

Based on the query above, the results will be displayed in the table below.

id	Employee name	Position title	total revenue all time	total hours worked	avg revenue per hour	avg revenue per sale day
419	Fleur van der Linden	Filiaalmanager	€ 9.745.763,49	96,5	€ 100.992,37	€ 5.337,22
16	Willem Brouwer	Filiaalmanager	€ 5.436.767,89	54,25	€ 100.216,92	€ 2.977,42
96	Greta de Jong	Assistent Manager	€ 9.785.360,59	104,75	€ 93.416,33	€ 5.358,91
173	Dirk Brouwer	Verkoopmedewerker	€ 8.161.053,80	89,25	€ 91.440,38	€ 4.469,36
70	Yara Strik	Verkoopmedewerker (parttime)	€ 5.481.763,80	61,5	€ 89.134,37	€ 3.002,06
215	Noa de Jong	Assistent Manager	€ 4.088.935,74	48,75	€ 83.875,60	€ 2.239,29
10	Hilde de Jong	Filiaalmanager	€ 5.420.679,88	77	€ 70.398,44	€ 2.968,61
83	Femke Molenaar	Assistent Manager	€ 5.465.095,00	82,5	€ 66.243,58	€ 2.992,93
460	Finn van Beek	Verkoopmedewerker	€ 4.075.739,96	62	€ 65.737,74	€ 2.232,06
310	Esther Wolters	Assistent Manager	€ 8.124.670,01	131	€ 62.020,38	€ 4.449,44
306	Hendrik Jacobs	Filiaalmanager	€ 8.140.596,59	134,75	€ 60.412,59	€ 4.458,16
121	Ruben Strik	Assistent Manager	€ 8.174.098,40	136,25	€ 59.993,38	€ 4.476,51
20	Sanne van der Meer	Filiaalmanager	€ 5.442.563,57	96,25	€ 56.546,12	€ 2.980,59
236	Lisa Peeters	Filiaalmanager	€ 4.879.129,32	86,5	€ 56.406,12	€ 2.672,03
188	Noa van der Linden	Verkoopmedewerker	€ 4.101.910,01	84	€ 48.832,26	€ 2.246,39
162	Esther Klein	Filiaalmanager	€ 8.131.050,28	168,5	€ 48.255,49	€ 4.452,93
119	Gerard Vos	Filiaalmanager	€ 8.174.198,44	174,25	€ 46.910,75	€ 4.476,56
196	Finn Smeets	Assistent Manager	€ 4.101.066,33	88	€ 46.603,03	€ 2.245,93
446	Lisa de Groot	Assistent Manager	€ 4.078.185,18	93	€ 43.851,45	€ 2.233,40
27	Ruben Bos	Verkoopmedewerker	€ 2.746.681,06	68,75	€ 39.951,72	€ 1.504,21
84	Adriana van Leeuwen	Filiaalmanager	€ 5.482.117,55	138,25	€ 39.653,65	€ 3.002,25
467	Finn van den Heuvel	Assistent Manager	€ 4.100.365,09	104	€ 39.426,59	€ 2.245,54

id	Employee name	Position title	total revenue all time	total hours worked	avg revenue per hour	avg revenue per sale day
341	Ingrid Groen	Assistent Manager	€ 4.867.566,08	123,5	€ 39.413,49	€ 2.665,70
435	Sanne Wolters	Verkoopmedewerker (parttime)	€ 4.082.335,90	109	€ 37.452,62	€ 2.235,67
179	Wilhelmina de Jong	Verkoopmedewerker	€ 4.072.539,17	118	€ 34.513,04	€ 2.230,31
375	Iris van den Berg	Verkoopmedewerker (parttime)	€ 4.863.214,60	141,5	€ 34.369,01	€ 2.663,32
219	Stijn van de Ven	Assistent Manager	€ 4.091.102,58	126	€ 32.469,07	€ 2.240,47
135	Finn Dekker	Filiaalmanager	€ 3.270.579,35	101,25	€ 32.302,02	€ 1.791,12
453	Hendrik Dijkstra	Verkoopmedewerker (parttime)	€ 4.086.302,73	129	€ 31.676,77	€ 2.237,84
191	Lynn Hermans	Verkoopmedewerker (parttime)	€ 4.076.042,15	129,25	€ 31.536,11	€ 2.232,22
33	Pieter Visser	Filiaalmanager	€ 2.769.748,82	90,75	€ 30.520,65	€ 1.516,84
54	Jan Prins	Verkoopmedewerker	€ 2.346.251,01	78	€ 30.080,14	€ 1.284,91
39	Gerard Jacobs	Assistent Manager	€ 2.751.831,33	91,75	€ 29.992,71	€ 1.507,03
52	Tessa van Beek	Verkoopmedewerker	€ 2.323.746,17	79,25	€ 29.321,72	€ 1.272,59
299	Wilhelmina de Haan	Assistent Manager	€ 3.252.005,14	111	€ 29.297,34	€ 1.780,94
293	Geertruida Kok	Assistent Manager	€ 3.263.790,70	114,25	€ 28.567,10	€ 1.787,40
63	Thomas Wolters	Verkoopmedewerker	€ 2.323.316,98	81,75	€ 28.419,78	€ 1.272,35
152	Willem de Boer	Verkoopmedewerker (parttime)	€ 2.707.782,34	95,75	€ 28.279,71	€ 1.482,90
398	Lieke Schouten	Verkoopmedewerker	€ 3.298.029,04	120,75	€ 27.312,87	€ 1.806,15
37	Liam van Vliet	Filiaalmanager	€ 2.744.239,11	102	€ 26.904,31	€ 1.502,87
267	Charlotte de Jong	Verkoopmedewerker (parttime)	€ 2.756.125,78	102,5	€ 26.889,03	€ 1.509,38
290	Thijs van den Heuvel	Filiaalmanager	€ 3.274.211,78	123,25	€ 26.565,61	€ 1.793,11
437	Emma Dekker	Verkoopmedewerker	€ 4.091.552,92	160,75	€ 25.452,90	€ 2.240,72
142	Kees Strik	Filiaalmanager	€ 2.699.906,15	106,25	€ 25.410,88	€ 1.478,59
251	Jesse Schouten	Assistent Manager	€ 2.714.564,99	107,25	€ 25.310,63	€ 1.486,62
348	Lars Vermeer	Filiaalmanager	€ 3.281.049,35	130	€ 25.238,84	€ 1.796,85
400	Femke van Dijk	Verkoopmedewerker (parttime)	€ 3.277.602,06	130,75	€ 25.067,70	€ 1.794,96

id	Employee name	Position title	total revenue all time	total hours worked	avg revenue per hour	avg revenue per sale day
132	Johanna Peeters	Verkoopmedewerker	€ 3.263.294,20	131,25	€ 24.863,19	€ 1.787,13
245	Esther Molenaar	Filiaalmanager	€ 2.726.282,39	110	€ 24.784,39	€ 1.493,04
281	Johanna Kok	Assistent Manager	€ 3.276.503,92	132,75	€ 24.681,76	€ 1.794,36
469	Max de Boer	Filiaalmanager	€ 4.056.502,89	170,5	€ 23.791,81	€ 2.221,52
477	Esther van Vliet	Filiaalmanager	€ 1.949.934,18	82,25	€ 23.707,41	€ 1.067,87
35	Niels Mulder	Assistent Manager	€ 2.735.668,99	122,75	€ 22.286,51	€ 1.498,18
134	Mila de Haan	Assistent Manager	€ 3.264.811,26	146,75	€ 22.247,44	€ 1.787,96
206	Thijs de Groot	Filiaalmanager	€ 4.120.296,25	187,75	€ 21.945,65	€ 2.256,46
468	Lynn de Jong	Verkoopmedewerker	€ 4.078.297,25	186	€ 21.926,33	€ 2.233,46
410	Joost Vos	Verkoopmedewerker (parttime)	€ 2.425.205,46	110,75	€ 21.898,02	€ 1.328,15
199	Yara Schouten	Verkoopmedewerker (parttime)	€ 2.454.594,46	113,25	€ 21.674,12	€ 1.344,25
222	Frans de Haan	Verkoopmedewerker (parttime)	€ 4.101.238,62	189,25	€ 21.671,01	€ 2.246,02
274	Hendrik Visser	Verkoopmedewerker (parttime)	€ 2.745.876,72	127,5	€ 21.536,29	€ 1.503,77
484	Yara Bos	Assistent Manager	€ 1.957.346,45	91	€ 21.509,30	€ 1.071,93
287	Petronella Jansen	Filiaalmanager	€ 3.267.663,40	155	€ 21.081,70	€ 1.789,52
276	Esther de Graaf	Verkoopmedewerker (parttime)	€ 2.745.733,79	131	€ 20.959,80	€ 1.503,69
356	Thijs Vermeer	Verkoopmedewerker (parttime)	€ 3.279.792,51	159,5	€ 20.562,96	€ 1.796,16
349	Emma van der Velden	Assistent Manager	€ 3.273.955,47	159,5	€ 20.526,37	€ 1.792,97
273	Liam van Vliet	Filiaalmanager	€ 2.763.891,57	137,25	€ 20.137,64	€ 1.513,63
253	Lotte Smit	Verkoopmedewerker (parttime)	€ 2.719.048,98	136,25	€ 19.956,32	€ 1.489,07
147	Evi de Jong	Filiaalmanager	€ 2.718.929,11	136,75	€ 19.882,48	€ 1.489,01
24	Cornelia Bakker	Assistent Manager	€ 2.735.511,48	138,25	€ 19.786,70	€ 1.498,09
248	Eva Mulder	Verkoopmedewerker (parttime)	€ 2.705.132,73	138	€ 19.602,41	€ 1.481,45
153	Adriana de Wit	Verkoopmedewerker (parttime)	€ 2.728.518,13	142,75	€ 19.113,96	€ 1.494,26
482	Gerard Hendriks	Assistent Manager	€ 1.963.479,46	103,25	€ 19.016,75	€ 1.075,29
143	Sara Strik	Verkoopmedewerker (parttime)	€ 2.723.660,83	144,25	€ 18.881,53	€ 1.491,60

id	Employee name	Position title	total revenue all time	total hours worked	avg revenue per hour	avg revenue per sale day
200	Lucas Hermans	Verkoopmedewerker (parttime)	€ 2.452.379,21	131,25	€ 18.684,79	€ 1.343,03
328	Wilhelmina de Wit	Verkoopmedewerker (parttime)	€ 1.646.976,77	88,75	€ 18.557,48	€ 901,96
58	Thijs Kok	Assistent Manager	€ 2.345.506,39	131,5	€ 17.836,55	€ 1.284,51
429	Jesse Janssen	Filiaalmanager	€ 1.633.369,19	91,75	€ 17.802,39	€ 894,51
145	Mila de Graaf	Assistent Manager	€ 2.696.561,46	151,75	€ 17.769,76	€ 1.476,76
362	Cornelia de Haan	Verkoopmedewerker (parttime)	€ 1.617.887,00	91,75	€ 17.633,65	€ 886,03
51	Stijn Meijer	Verkoopmedewerker	€ 2.324.545,17	135	€ 17.218,85	€ 1.273,03
270	Levi Jansen	Filiaalmanager	€ 2.741.978,53	161,25	€ 17.004,52	€ 1.501,63
255	Lisa van den Heuvel	Verkoopmedewerker (parttime)	€ 2.704.811,18	166	€ 16.294,04	€ 1.481,28
280	Liam Klein	Filiaalmanager	€ 2.735.181,19	169,75	€ 16.113,00	€ 1.497,91
399	Jesse Molenaar	Verkoopmedewerker	€ 3.271.487,42	204,25	€ 16.017,07	€ 1.791,61
68	Bram Wolters	Filiaalmanager	€ 2.321.354,93	150,25	€ 15.449,95	€ 1.271,28
327	Noah van Vliet	Verkoopmedewerker (parttime)	€ 1.642.681,08	113,75	€ 14.441,15	€ 899,61
249	Daan Hermans	Assistent Manager	€ 2.706.493,15	189,75	€ 14.263,47	€ 1.482,20
56	Thomas Hoekstra	Verkoopmedewerker (parttime)	€ 2.342.468,81	164,25	€ 14.261,61	€ 1.282,84
486	Sophie van der Velden	Verkoopmedewerker	€ 1.966.162,65	143,5	€ 13.701,48	€ 1.076,76
366	Sem de Boer	Filiaalmanager	€ 1.629.496,94	120,75	€ 13.494,80	€ 892,39
41	Gerard van Dijk	Verkoopmedewerker	€ 1.225.718,83	100	€ 12.257,19	€ 671,26
476	Charlotte van Dijk	Verkoopmedewerker (parttime)	€ 1.964.375,21	163,5	€ 12.014,53	€ 1.075,78
365	Lisa de Haan	Filiaalmanager	€ 1.635.279,21	140,5	€ 11.639,00	€ 895,55
411	Pieter Hoekstra	Assistent Manager	€ 2.439.957,85	220	€ 11.090,72	€ 1.336,23
43	Lars de Haan	Filiaalmanager	€ 1.225.905,75	123	€ 9.966,71	€ 671,36
427	Lars Willems	Verkoopmedewerker	€ 1.625.246,03	163,25	€ 9.955,57	€ 890,06
433	Eva Jacobs	Verkoopmedewerker (parttime)	€ 1.636.122,04	171	€ 9.567,97	€ 896,01
371	Niels Janssen	Verkoopmedewerker (parttime)	€ 1.226.865,86	134,5	€ 9.121,68	€ 671,89
44	Fenna Peters	Assistent Manager	€ 1.217.847,63	141	€ 8.637,22	€ 666,95

id	Employee name	Position title	total revenue all time	total hours worked	avg revenue per hour	avg revenue per sale day
322	Levi van der Linden	Assistent Manager	€ 1.633.625,20	189,75	€ 8.609,36	€ 894,65
373	Hans Vermeer	Filiaalmanager	€ 1.214.682,31	156	€ 7.786,43	€ 665,21
374	Geertruida Dijkstra	Filiaalmanager	€ 1.205.958,90	159,25	€ 7.572,74	€ 660,44
367	Emma de Graaf	Verkoopmedewerker	€ 1.224.098,43	180,75	€ 6.772,33	€ 670,37
46	Bram de Graaf	Verkoopmedewerker (parttime)	€ 1.214.430,05	193,25	€ 6.284,24	€ 665,08

5. How many customers purchase more than 3 times per month?

To answer this question, we use the orders table. The orders table is used to identify customer transactions, filter only completed orders (Voltooid), and extract the transaction date to group data by year and month.

```

SELECT
    year,
    month,
    COUNT(*) AS loyal_customers,
    ROUND(AVG(total_purchases), 2) AS avg_purchases_per_customer
FROM (
    SELECT
        o.customer_id,
        YEAR(o.order_datetime) AS year,
        MONTH(o.order_datetime) AS month,
        COUNT(*) AS total_purchases
    FROM orders o
    WHERE o.customer_id IS NOT NULL
        AND o.order_status = 'Voltooid'
    GROUP BY
        o.customer_id,
        YEAR(o.order_datetime),
        MONTH(o.order_datetime)
    HAVING COUNT(*) > 3
) AS monthly_customer_freq
GROUP BY year, month
ORDER BY year, month;

```

Based on the query above, the results will be displayed in the table below.

year	month	loyal customers	avg purchases/customer
2019	1	5.219	4,25
2019	2	3.571	4,23
2019	3	5.253	4,25
2019	4	4.473	4,25
2019	5	5.084	4,25
2019	6	4.513	4,26
2019	7	5.072	4,25
2019	8	5.561	4,27
2019	9	4.179	4,24
2019	10	5.221	4,25
2019	11	4.799	4,26

year	month	loyal customers	avg purchases/customer
2019	12	4.883	4,25
2020	1	4.954	4,26
2020	2	4.215	4,25
2020	3	4.813	4,23
2020	4	4.688	4,24
2020	5	5.126	4,26
2020	6	4.550	4,24
2020	7	4.961	4,26
2020	8	5.076	4,26
2020	9	4.517	4,25
2020	10	5.411	4,26
2020	11	4.303	4,24
2020	12	5.141	4,26
2021	1	4.956	4,24
2021	2	3.559	4,22
2021	3	4.946	4,26
2021	4	4.480	4,24
2021	5	4.860	4,26
2021	6	4.701	4,25
2021	7	5.209	4,26
2021	8	4.834	4,25
2021	9	4.687	4,25
2021	10	5.002	4,26
2021	11	4.384	4,24
2021	12	5.016	4,25
2022	1	4.994	4,25
2022	2	3.855	4,23
2022	3	5.093	4,26
2022	4	4.815	4,25
2022	5	4.738	4,24
2022	6	4.515	4,25
2022	7	5.070	4,27
2022	8	4.986	4,24
2022	9	4.677	4,25
2022	10	4.985	4,26
2022	11	4.615	4,24
2022	12	5.426	4,26
2023	1	4.619	4,26
2023	2	3.703	4,22
2023	3	5.091	4,26
2023	4	4.466	4,25
2023	5	5.031	4,25
2023	6	4.614	4,26
2023	7	4.819	4,25
2023	8	5.149	4,25

year	month	loyal customers	avg purchases/customer
2023	9	4.941	4,26
2023	10	4.704	4,24
2023	11	4.857	4,25
2023	12	5.022	4,27